

RECURSION

Day 72
20/09/2025

Mathematics $\leftrightarrow$ Recursion Relation

① Recurrence Relations $\rightarrow$ Recursive functions

- In Math many problems are defined as recurrence relations
A formula that Expresses the n^{th} term using previous terms
- In Programming, recurrence relations are implemented naturally with recursive functions

Example : Factorial

In Mathematics

$$n! = \begin{cases} 1 & \text{if } n=0 \\ n \cdot (n-1)! & \text{if } n > 0 \end{cases}$$

In Java programming

```
int factorial (int n) {  
    if (n==0) return 1;  
    return n * factorial (n-1);  
}
```

② Mathematical Induction $\leftrightarrow$ Proof of Recursion Correctness

- Recursion looks like induction proofs
 - ① Base case = Induction Base step
 - ② Recursive step = induction hypothesis + Inductive step

Example : Factorial again

- Base case $0! = 1$
- Recursive step : Assume $(n-1)!$ is correct then $n! = n \cdot (n-1)!$
- This Matches How exactly recursion Executes

③ Mathematical Sequence via Recursion

- Many Mathematical Sequence are recursive in nature

Example : Fibonacci Sequence

0 1 1 2 3 5 8

$$F(n) = \begin{cases} 0 & \text{if } n=0 \\ 1 & \text{if } n=1 \\ F(n-1) + F(n-2) & \text{if } n > 1 \end{cases}$$

Recursive Code

```
int fib(int n) {  
    if (n == 0) return 0;  
    if (n == 1) return 1;  
    return fib(n-1) + fib(n-2);  
}
```

④ Recursive Mathematical functions

Some Mathematical functions are inherently recursive

- GCD (Euclidean Algorithm)

$$\text{gcd}(a, b) = \begin{cases} a & \text{if } b=0 \\ \text{gcd}(b, a \bmod b) & \text{if } b > 0 \end{cases}$$

Java

```
int gcd (int a, int b) {  
    if (b == 0) return a;  
    return (b, a % b);  
}
```

}

• Ackermann Function (a famous example of deep recursion):

$$A(m, n) = \begin{cases} n + 1 & \text{if } m = 0 \\ A(m - 1, 1) & \text{if } m > 0, n = 0 \\ A(m - 1, A(m, n - 1)) & \text{if } m > 0, n > 0 \end{cases}$$

⑤ Recursion and Infinite Series

Even infinite Sums / products are recursive

Example Power function a^n

$$a^n = \begin{cases} 1 & \text{if } n = 0 \\ a \cdot a^{n-1} & \text{if } n > 0 \end{cases}$$

Recursive code

```
int power (int a, int n) {  
    if (n == 0) return 1;  
    return a * power(a, n - 1);  
}
```

}

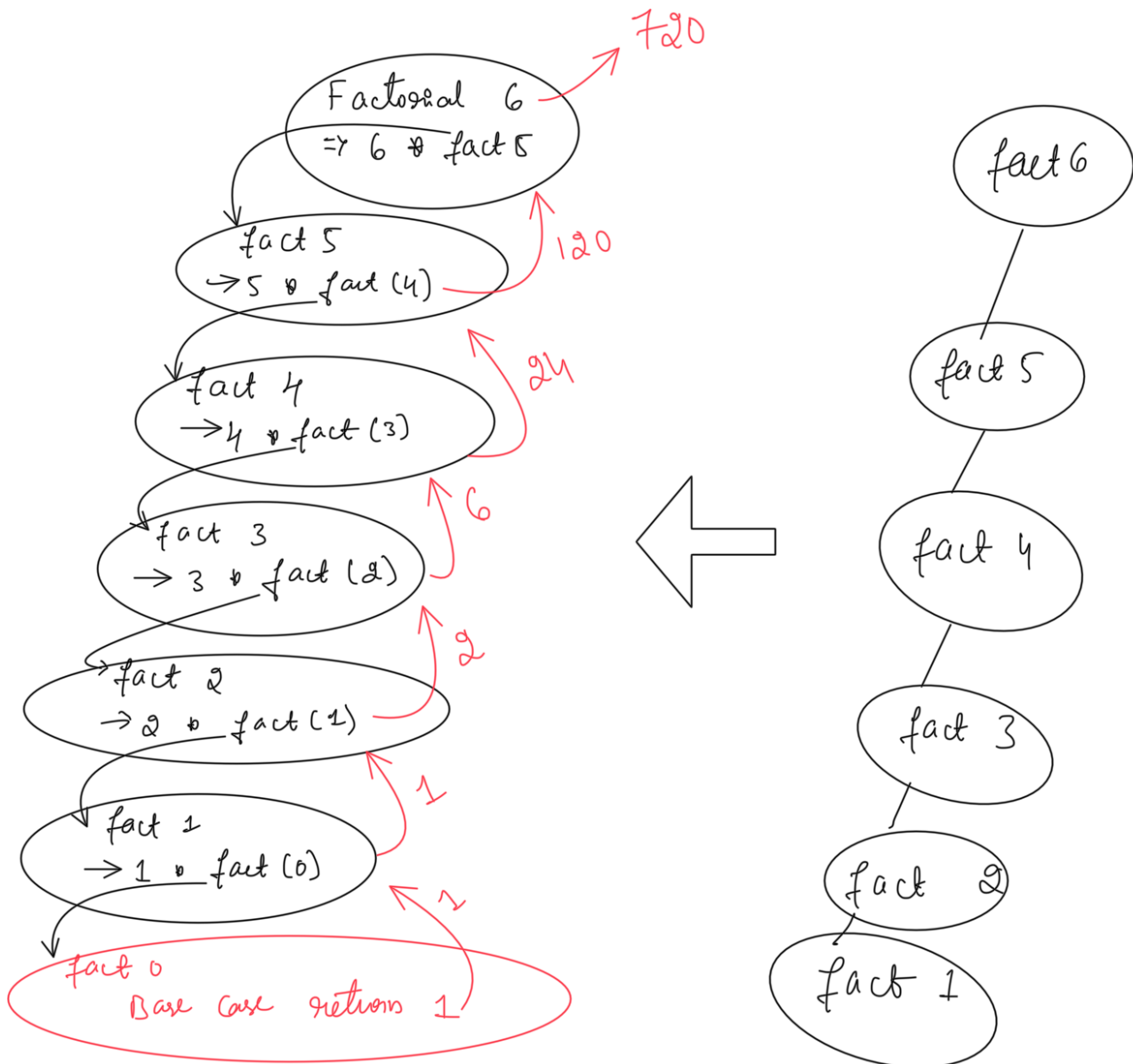
Why this Connection Matters

- Mathematics defines recursion formally (via recurrence relations & induction)
- Programming implements it via recursive functions
- Many algorithms (sorting, graph traversal, DP) are Based on Mathematical recurrence relations

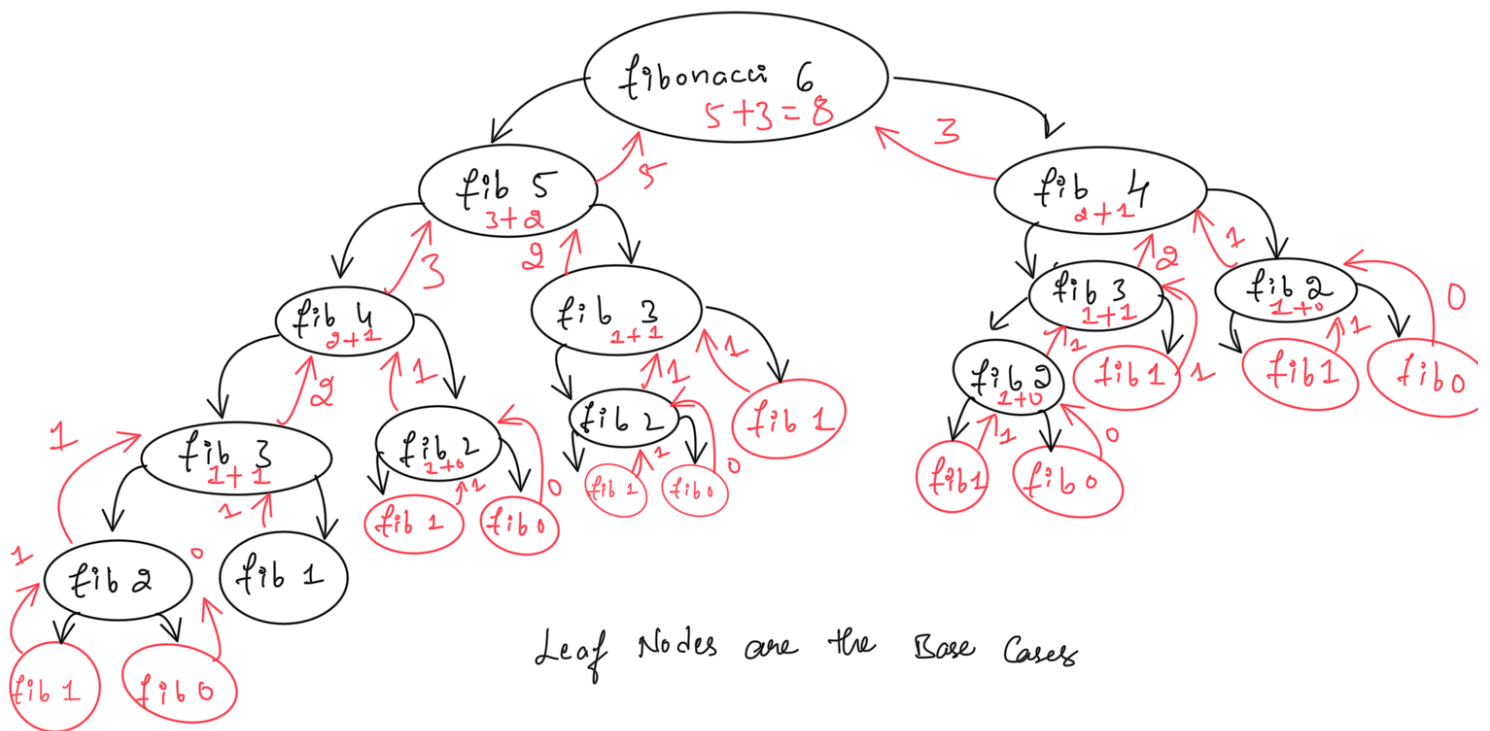
Recursion Tree

The Recursion tree helps to visualize the recursive function calls

Factorial Recursion Tree



Fibonacci Recursion Tree



Each time Two Recursive calls are called except Base Cases